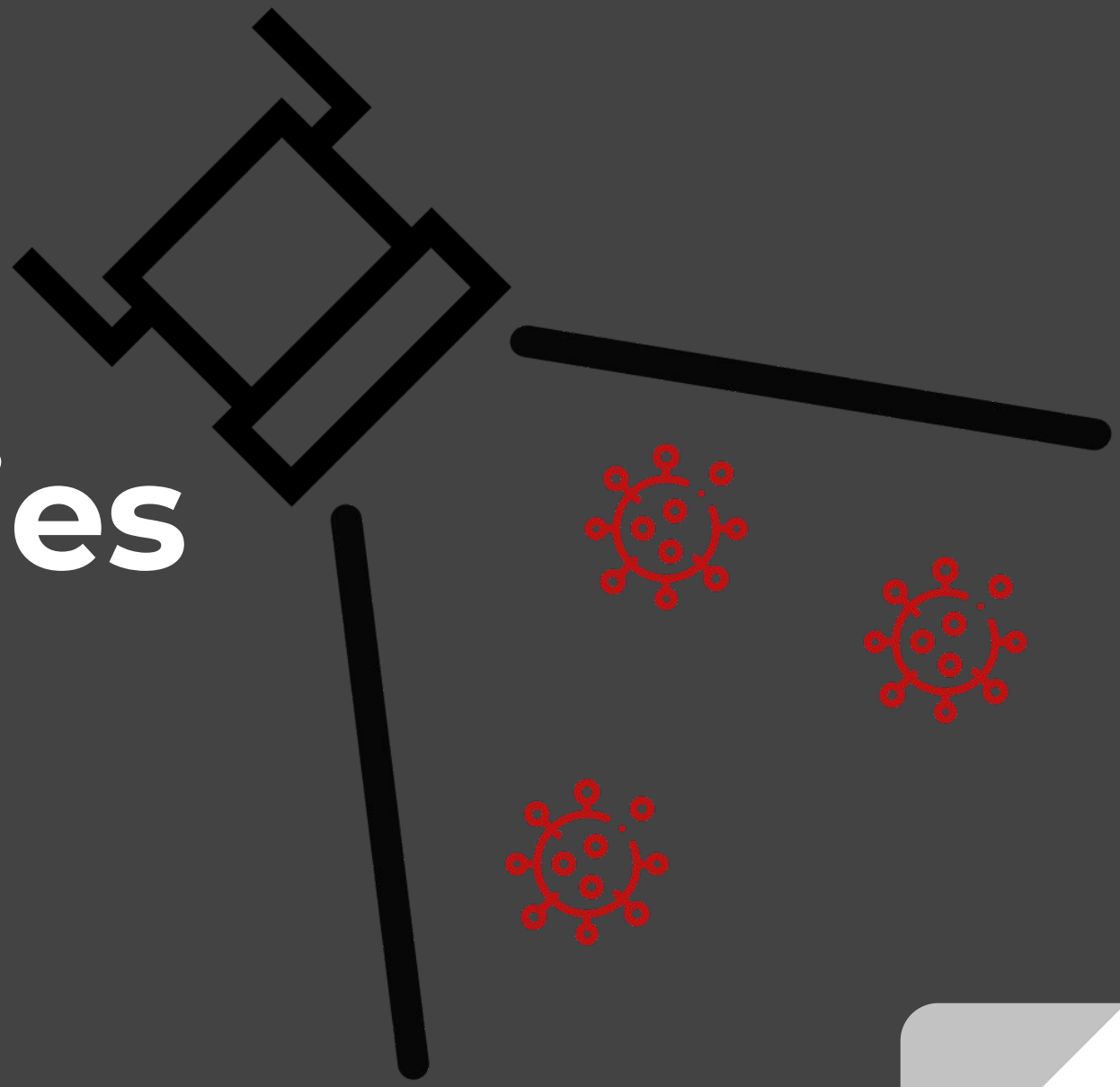
FloodLAMP Biotechnologies

a Public Benefit Corporation



floodlamp.bio

Our Mission is to deploy mass screening to stop the spread of infectious disease

Overview of Technology Involved

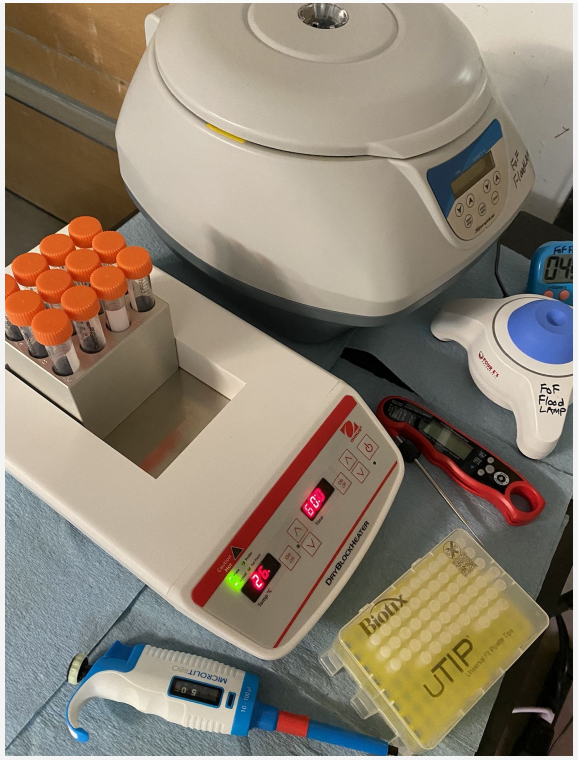
Sample Collection

- **Coordinated by Custom App**
- **Self pooling by household**

Assay Chemistry - Rabe Cepko Protocol from Harvard Medical School:

- **Chemical Inactivation using TCEP/EDTA**
- **Nucleic Acid Purification Protocol using ultra cheap Silica**
- **Isothermal LAMP amplification with colorimetric readout**

Rabe Cepko RT-LAMP Assay



Inactivation:

Chemical (TCEP) + Heat
Deployable outside a lab

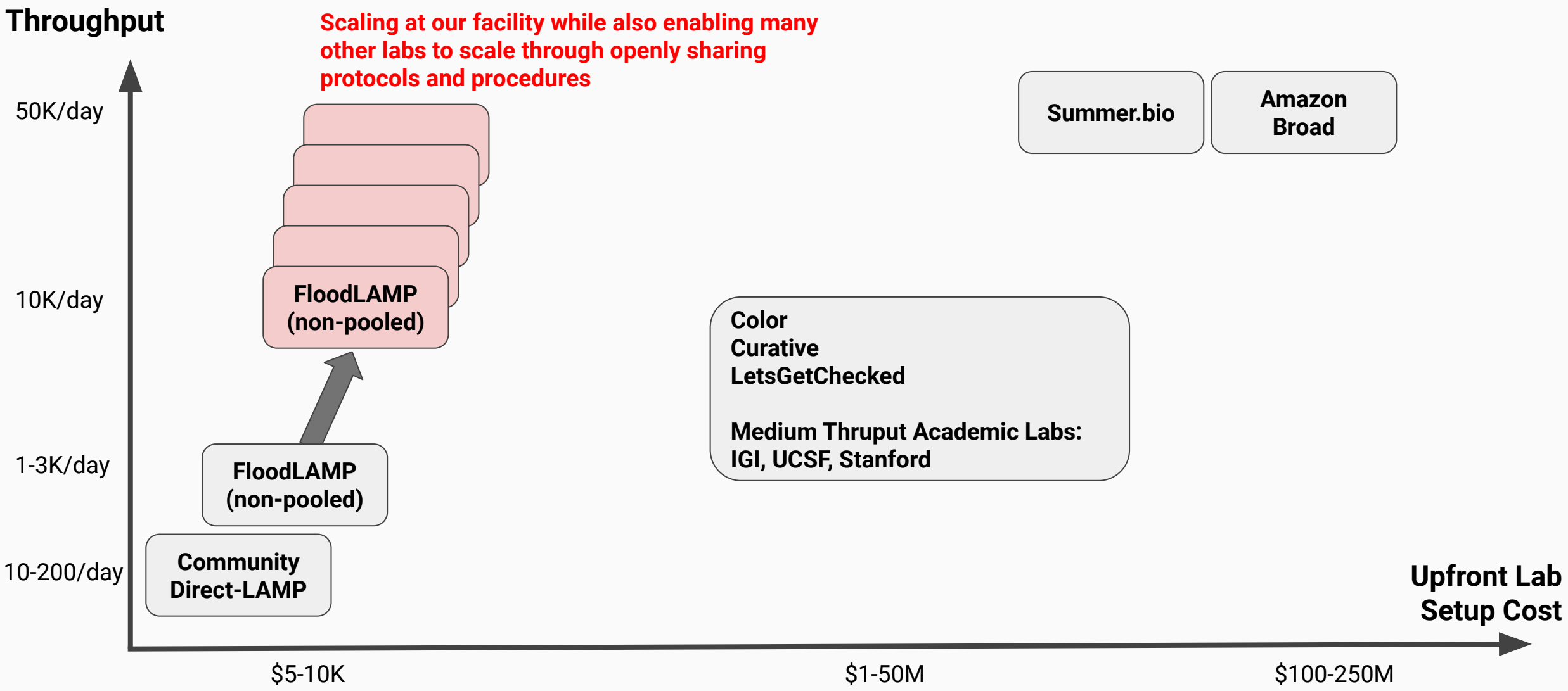
Purification:

Bulk Silica
Allows larger volumes
Ultra cheap

Amplification:

Colorimetric LAMP
Isothermal
Instrument Free

Unique Positioning



Scaling in Multiple Ways

1) Increase capacity at our facility with automation

OPENCOVID19 INITIATIVE

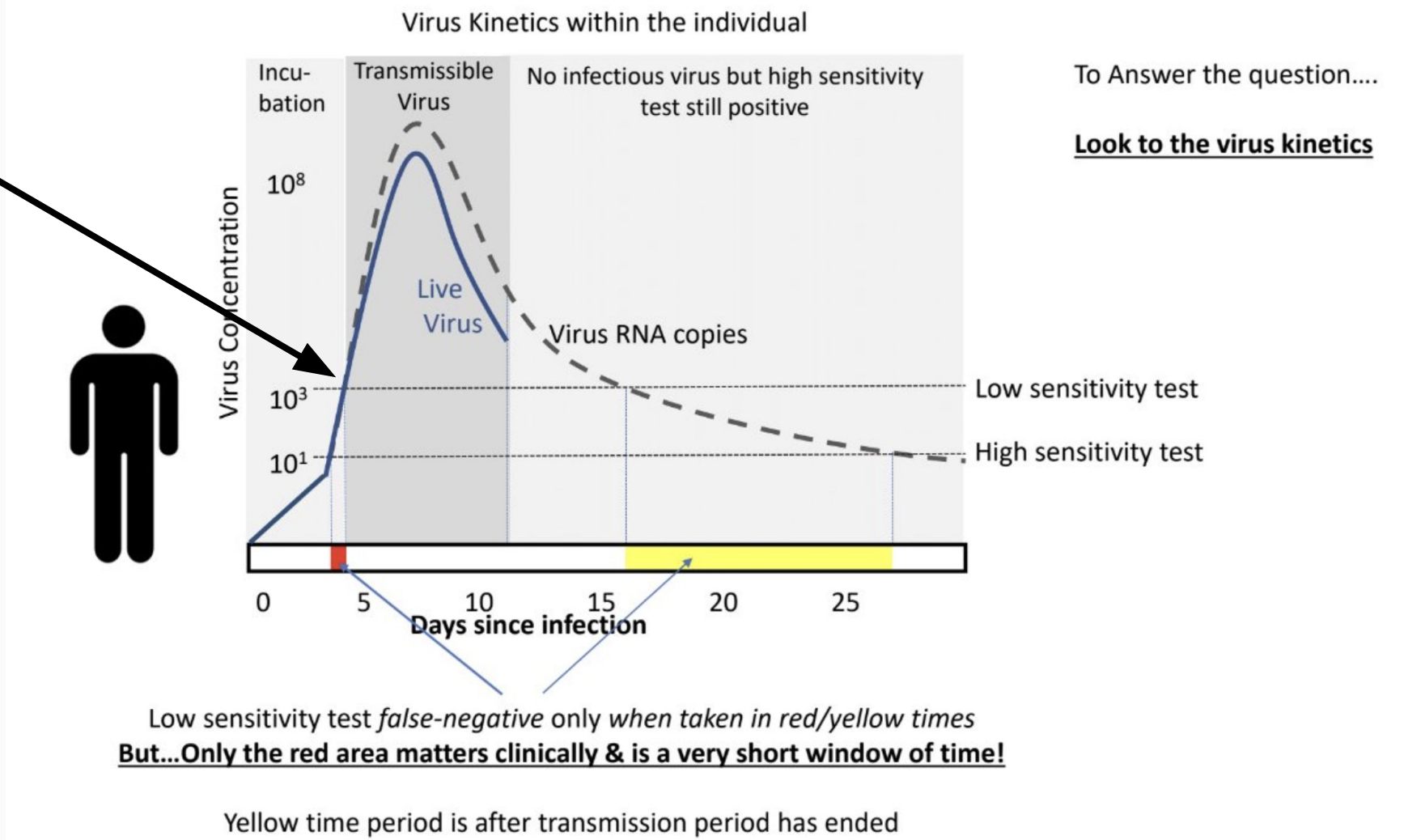
2) Enable distributed network of labs through open protocols

3) IVD EUA to widely distribute full assay kits

Pooling for Infectious Detection

Threshold for "Capable to Infect Others" ~ 1e6/ml

Pooling uses assay sensitivity to scale up coverage - and detect unknown infectious people in interacting populations



To Answer the question....

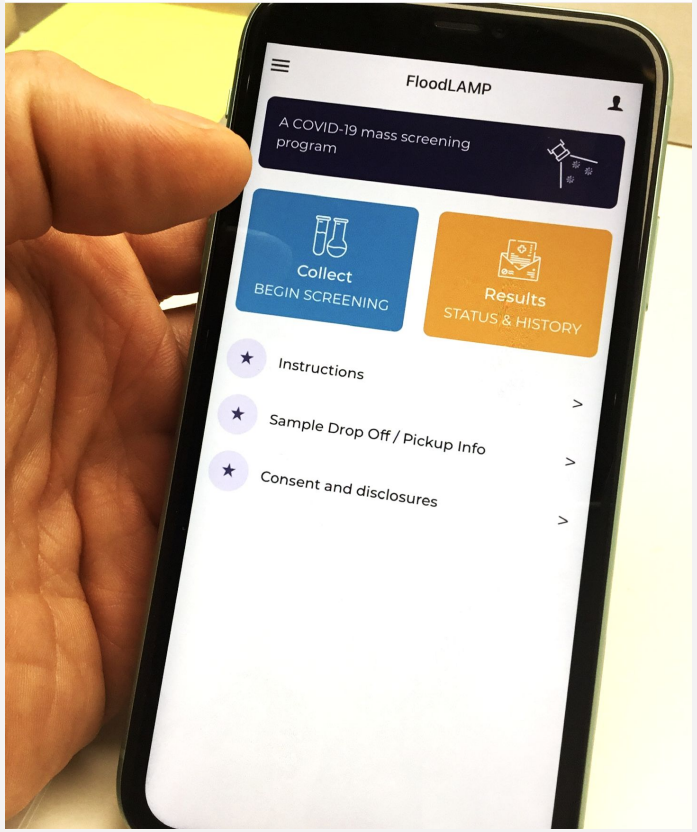
Look to the virus kinetics

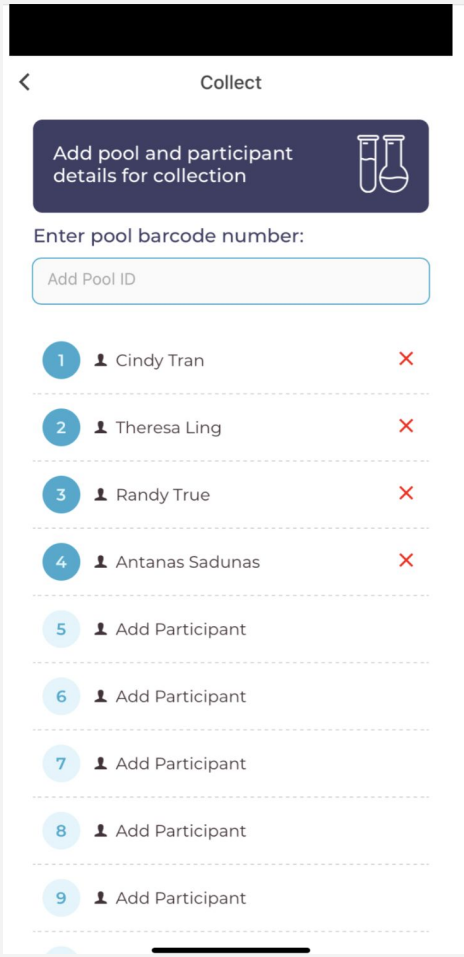
Benefitting Public Health

Our open protocol and scaling-friendly set up costs mean low overhead (training, cost, equipment) to better screen the populations that need it most.

User-centered Screening

- **At-home collection**
- **Pooling by household or pod**
- **Custom software for recording samples and tracking results.**
- **Prompts for repeat screens**



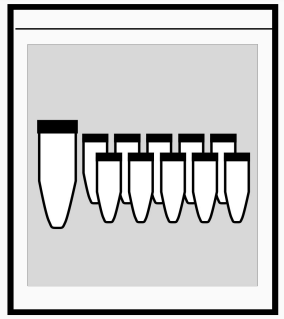


Sample Collection Workflow

1 Register

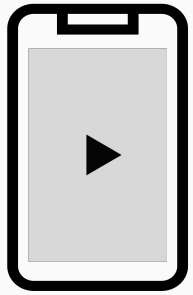
- Visit app and register
- Your info added
- Your organization (if applicable)

2 Get



- Receive your kit in the mail
- Review instructions

3 Learn

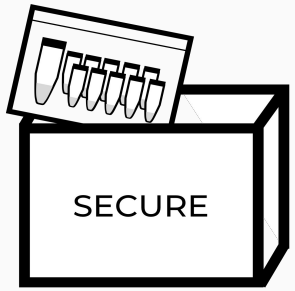


- Watch videos on collection techniques and process
- Prepare other participants if you're a sponsor

4 Collect

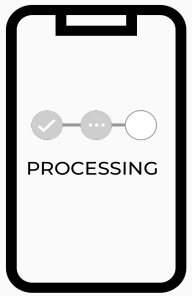
- Identify who you're collecting for in app
- Collect individual samples + record in app
- pool samples + record in app

5 Drop off



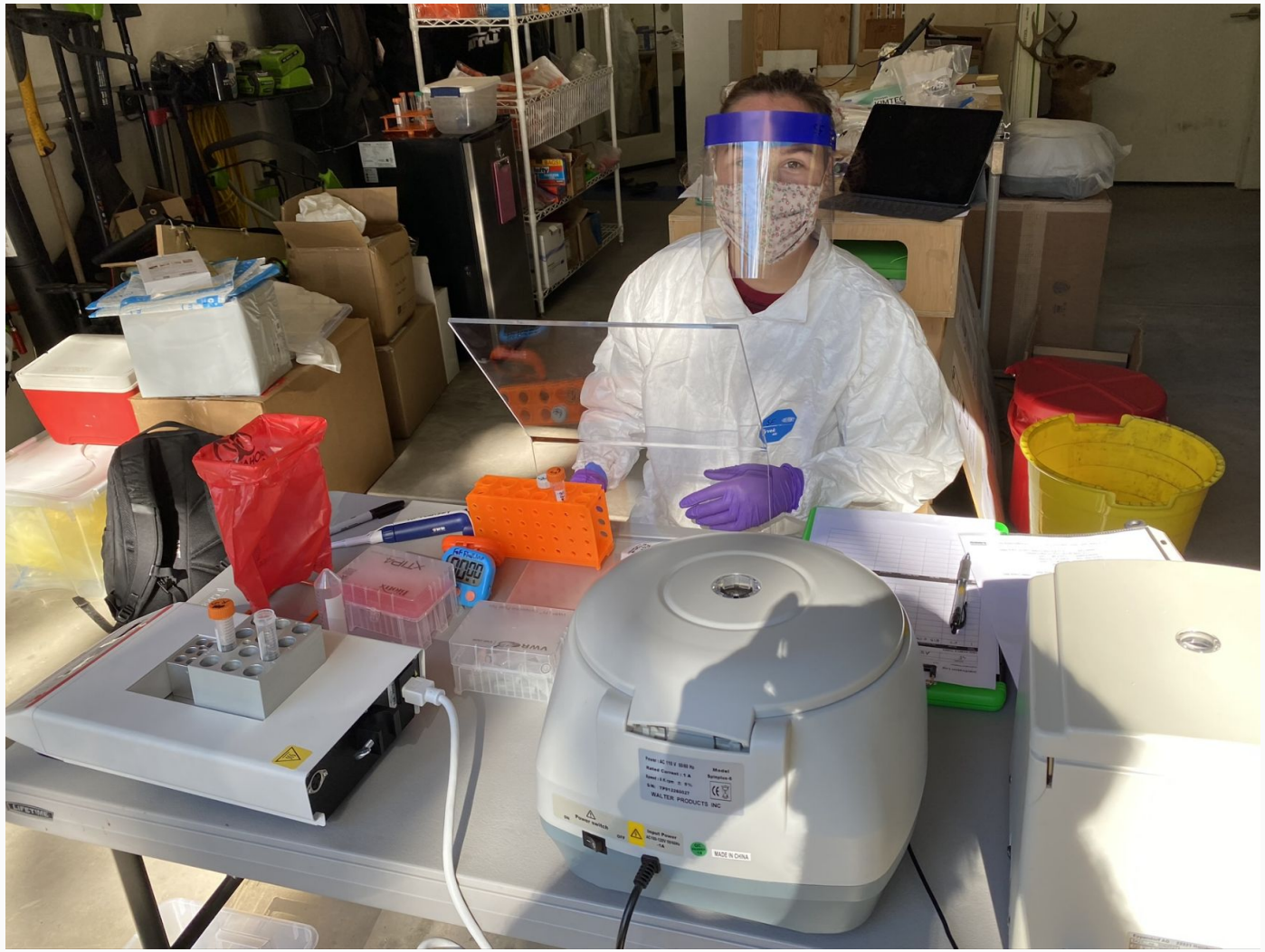
- Drop samples in secure drop box and record in app OR
- Drop with staff member who records in system

6 View status



- See state of your sample (in transit, processing, result, etc)
- See updates for next collection
- Sharing options for adoption spread

Field Inactivation Station



FloodLAMP Facilities at MBC Biolabs